Order code

Monitor unit for Dew Star series (S-1, S-2, S-3 common)

Order Code	Specification
S-1M-0	SI unit

Main Unit

S-1series for normal humidity *1

Order Code	Specification		
	Material of mirror	Sensor coating	Heater *2
S-1S-0	Rhodium	NO	NO
S-1S-4	Platinum	NO	NO
S-1S-6	Platinum	YES	NO
S-1S-8	Rhodium	NO	YES
S-1S-C	Platinum	NO	YES
S-1S-E	Platinum	YES	YES

S-2 series for low humidity*1

Order Code	Specification	
	Material of mirror	Sensor coating
S-2S-0	Rhodium	NO
S-2S-4	Platinum	NO
S-2S-6	Platinum	YES

S-3 series for high humidity*1

Order Code	Specification	
	Material of mirror	Sensor coating
S-3S-0	Rhodium	NO
S-3S-4	Platinum	NO
S-3S-6	Platinum	YES

Options

Order Code	Specification
S-1S-001	Temperature sensor (Pt100Ω 4-wire type)
S-1S-002	Pressure sensor (4 ～ 20mA, 2-wire type)
AU-3	Air pump unit (100VAC 50/60Hz)

*1 Monitoring display S-1M-0 is needed.
*2 Needed when dew point is higher than room temperature. Integrated with sensor and controlling unit. Must be ordered before shipped out of our factory. External temperature controller needed separately.

Calibration & Repair service

SHINYEI provides calibration service traceable to National Metrology Institute of Japan (NMIJ) where standardizes humidity in Japan.
SHINYEI also operates all repairs of DewStar at its own factory with fast lead time.



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DewStar Series
Chilled Mirror Dew Point Hygrometers

SHINYEI makes your lab HUMIDITY ARTIST.

Shinyei field-proven chilled mirror technology

Wide dew point range covered
Auto mirror cleaning function (ACCU-STAR)
Corrosive gas measurement available* Sulfur Oxides (SOx), Nitrogen Oxides (NOx), Bromine (Br), Chlorine(Cl)
Psychrometric calculation function (Relative humidity, Absolute humidity, Moisture percentage etc.)



Chilled Mirror Dew Point Hygrometers

Lineup, Applications, Features

Multifunction and high accuracy enable various applications

95℃
50℃
25℃
15℃
0℃
-40℃
-65℃
-75℃

For high humidity

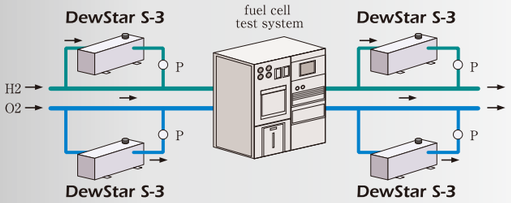
Chilled Mirror Dew Point Hygrometers S-3

0℃DP ～ 95℃DP

-Standard reference for electronic component durability test
-Dew point control for firing furnaces
-Humidity control for thermo-hygrostat chamber
-Dew point control for fuel cell test system







For normal humidity

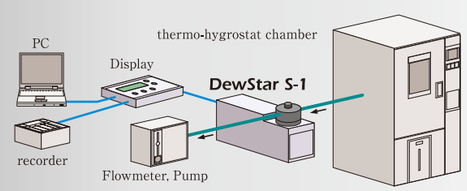
Chilled Mirror Dew Point Hygrometers S-1

-40℃DP ～ 50℃DP

-Standard reference for calibration
-Moisture control for ceramic firing furnaces
-Humidity control for thermo-hygrostat chamber









Display can be combined with body as one

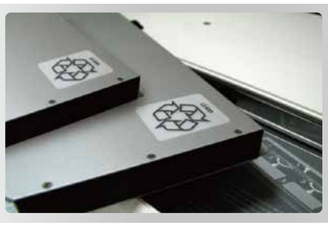
For low humidity

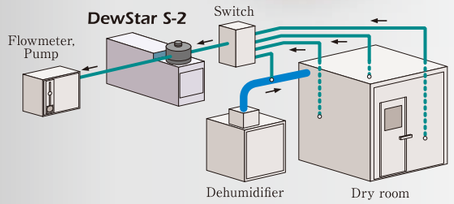
Chilled Mirror Dew Point Hygrometers S-2

-65℃DP ～ 15℃DP (Air cooling)
-75℃DP ～ 15℃DP (water cooling)

-Monitoring dried air pipe at a production
-Moisture control at drying process
-Production process of lithium ion battery
-Measurement of gas with low dew point







Specification

	DewStar S-1	DewStar S-2	DewStar S-3
Operating range ≦95%rh	0 ～ 60℃ (sensor) 0 ～ 35℃ (monitor)	0 ～ 35℃ (sensor) 0 ～ 35℃ (monitor)	0 ～ 50℃ (sensor) 0 ～ 35℃ (monitor)
Measurement range of dew point (at 23℃)	-40℃DP ～ +50℃DP	-75℃DP ～ +15℃DP(Water-cooling) -65℃DP ～ +15℃DP(Air-cooling)	0℃DP～+95℃DP
Cooling capacity	-65℃	-95℃	-10℃
Operating pressure range	0.5MPaG	0.5MPaG	Barometric Pressure ～ 0.35MPaG
Reproducibility	±0.05℃DP	±0.1℃DP	±0.05℃DP
Sampling flow rate	0.5 ～ 2.5 L/min	0.5 ～ 2.5 L/min	0.5 ～ 2.5 L/min
System accuracy	±0.1℃DP (at 23℃DP)	±0.2℃DP (> -50℃DP) ±0.4℃DP (≦ -50℃DP)	±0.2℃DP
DisplayLCD	LCD (16 letters x 2 lines)	LCD (16 letters x 2 lines)	LCD (16 letters x 2 lines)
Measurement	Dew/frost point (℃), Temperature (℃), Air pressure (kPa)		
Psychrometric calculation	Relative humidity (%rh), Absolute humidity (g/m3), Moisture percentage (PPMv, PPMw)		
Temperature sensor (option)	Pt100Ω, 4-wire type		
Pressure sensor (option)	4 ～ 20mA, 2-wire type		
Analog output (at Display)	4 ～ 20mA, 0 ～ 5VDC (2ch, simultaneous usage available)		
Alarm output (at sensor)	PhotoMOS relay		
Digital output	RS232C (Measurement or Calculation value)		
Cable between main unit & monitor unit	Optimized cable (standard length 3m)		
Sample gas IN-OUT	IN-OUT 1/4 inch compression fitting		
Supply voltage	AC90 ～ 240V 50/60Hz		
Weight (Monitor unit)	500g	500g	500g
Weight (Main unit)	4kg	10kg	4kg
Power consumption	50W	200W	100W

Options

Temperature sensor S-1S-001

●Sensor: Pt100Ω 4-wire type ●Measurement range: -50 ～ +100℃
●Accuracy: ±0.2℃ (0℃) ●Dimension: φ3.2 x L250mm
●Cable length: 3m



Air pressure sensor

●Input voltage: 9 ～ 30VDC
●Accuracy: ±0.15%FS
●Output voltage: 4 ～ 20mA
●Pressure range: 0 ～ 210kPa
●Cable length: 3m
●Dimension / Weight: φ30 x 87mm / 200g



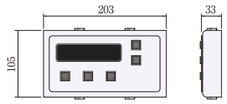
Air pump unit AU-3

Pumping sampling gas into Dew Star on measurement. Flow meter integrated.

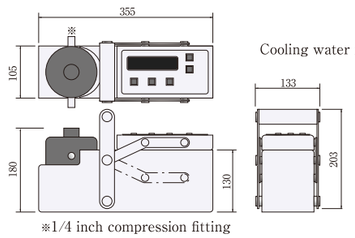


Dimension

●Monitor unit
(S-1, S-2, S-3 common)

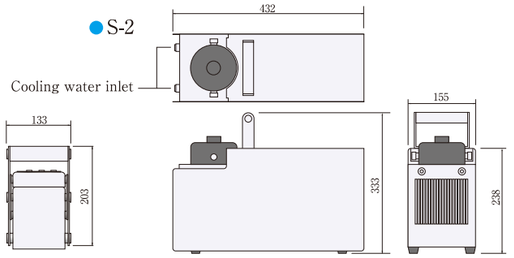


●S-1

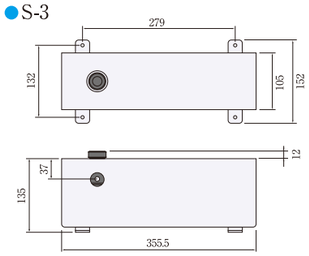


※1/4 inch compression fitting

●S-2



●S-3



What is the chilled mirror method

Optical condensation hygrometry is a precise technique for determining the water vapor content in gases by directly measuring dew point or frost temperatures. Using this technique, a metal mirror is cooled until it reaches a temperature at which a thin layer of condensation begins to form on it.
The dew layer is detected optically, and the mirror is held at that temperature. The mirror temperature, measured with a PRT, is an accurate indicator of the dew or frost point. Because these hygrometers are so accurate, they are used as a standard in many of the world's metrology laboratories.

